CloudSense AI

Intelligent Cloud Cost Optimization, Autonomous Remediation, and Denial-of-Wallet Mitigation using Machine Learning

Domains: Artificial Intelligence (AI) • Machine Learning (ML) • Cloud Computing • FinOps • Cloud Security (SecOps) • Explainable AI (XAI)

Alternative IEEE Paper Titles

1. **CloudSense AI: Intelligent Cloud Cost Optimization, Autonomous Remediation, and Denial-of-Wallet Mitigation using Explainable AI**
2. **An Autonomous FinOps and Cloud Security Framework using Machine Learning and Time-Series Forecasting**
3. **Predictive Cloud Resource Optimization and Intelligent Cloud Remediation using Machine Learning, Model Context Protocol (MCP), and AWS SDK (Boto3)**
4. **Mitigating Denial-of-Wallet (DoW) Attacks and Forecasting Cloud Unit Economics using Isolation Forest and XGBoost**

Project Overview

CloudSense AI is an AI-powered cloud optimization platform designed to help organizations reduce cloud expenditure while improving operational efficiency, security, and resource utilization. Modern cloud environments often contain idle resources, oversized infrastructure, and unpredictable workload patterns that significantly increase operational costs. CloudSense AI continuously monitors cloud infrastructure, analyzes historical and real-time metrics, and provides intelligent recommendations to optimize cloud resources before unnecessary expenses accumulate.

Unlike traditional cloud monitoring platforms that only display historical dashboards and alerts, CloudSense AI combines Machine Learning, Explainable AI (XAI), cloud analytics, and

Model Context Protocol (MCP) to build an intelligent decision-support platform. The system predicts future cloud spending, identifies inefficient resource utilization, detects abnormal cost behavior, and explains every recommendation using interpretable AI models.

One of the major strengths of the platform is its intelligent remediation workflow. Instead of generating Infrastructure-as-Code templates, CloudSense AI communicates with cloud services through the Model Context Protocol (MCP). After receiving user approval, the system securely invokes AWS services using the AWS SDK (Boto3) to perform actions such as stopping idle EC2 instances, resizing virtual machines, deleting unattached storage volumes, cleaning obsolete snapshots, configuring Auto Scaling Groups, and executing other optimization tasks. This approach provides enterprise-grade automation while maintaining complete human control over production environments.

CloudSense AI bridges FinOps, DevOps, and SecOps into a unified intelligent platform capable of forecasting cloud costs, protecting organizations from financial security threats such as Denial-of-Wallet (DoW) attacks, and delivering explainable optimization recommendations that improve operational confidence and business decision-making.

Problem Statement

Cloud computing enables organizations to rapidly deploy scalable applications, but uncontrolled cloud growth frequently leads to excessive operational costs and inefficient infrastructure utilization. Many enterprises allocate resources based on peak demand and later forget to optimize them, resulting in continuously running virtual machines, oversized compute instances, unattached storage volumes, idle databases, unused snapshots, and redundant cloud services that generate unnecessary expenses every month.

Existing cloud management platforms mainly provide dashboards and alerts without offering intelligent decision support or automated optimization capabilities. Engineers are often required to manually analyze cloud metrics, determine the root cause of high spending, identify the affected resources, estimate potential savings, and perform remediation actions manually. This process is time-consuming, error-prone, and unsuitable for rapidly changing cloud environments.

Another emerging challenge is the Denial-of-Wallet (DoW) attack, where attackers intentionally consume cloud resources to increase infrastructure costs rather than directly disrupting service availability. Traditional security tools primarily focus on availability-based attacks and frequently overlook financial attacks that silently increase cloud expenditure.

Organizations therefore require an intelligent cloud optimization system capable of continuously forecasting future costs, detecting inefficient resource utilization, identifying financial security threats, providing explainable AI recommendations, and safely executing approved remediation actions without compromising operational governance.

Proposed Solution

CloudSense AI introduces an AI-driven cloud intelligence platform that combines predictive analytics, anomaly detection, explainable machine learning, intelligent remediation, and enterprise cloud orchestration into a single integrated solution.

The platform continuously collects cloud billing data, infrastructure metrics, resource utilization statistics, application telemetry, and business Key Performance Indicators (KPIs). This information is processed through multiple Machine Learning models that forecast future cloud expenditure, identify abnormal spending behavior, detect underutilized infrastructure, and calculate business-aware Unit Economics.

Every recommendation generated by the AI engine includes a detailed explanation describing why the recommendation was produced, the confidence score of the prediction, the estimated monthly and annual savings, the associated business impact, and the potential operational risk. These explanations help engineers understand and validate optimization decisions before implementation.

Once a recommendation has been reviewed and approved, CloudSense AI communicates with cloud services through the Model Context Protocol (MCP). The MCP layer securely connects the AI platform with AWS services, while AWS SDK (Boto3) performs the required cloud operations. Supported remediation actions include starting or stopping EC2 instances, resizing compute resources, deleting unattached EBS volumes, removing obsolete snapshots, modifying Auto Scaling Groups, managing storage classes, and invoking AWS Lambda functions where necessary.

The platform also integrates with collaboration tools such as Slack, Microsoft Teams, Discord, Email, and custom webhook services through MCP-compatible connectors. Engineering teams receive real-time notifications whenever abnormal cloud spending, Denial-of-Wallet attacks, or optimization opportunities are detected, allowing rapid decision-making and secure cloud governance.

The overall objective of CloudSense AI is to transform traditional cloud monitoring into an intelligent AI-assisted FinOps platform capable of predicting, explaining, securing, and optimizing cloud infrastructure through modern AI technologies and enterprise automation.

Core Features

CloudSense AI consists of eight intelligent modules that work together to predict cloud expenditure, optimize infrastructure, improve cloud security, and automate remediation while maintaining enterprise governance. Each module addresses a specific challenge faced by organizations managing cloud infrastructure.

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1. Intelligent Cloud Cost Forecasting

CloudSense AI continuously predicts future cloud expenditure using historical billing data, infrastructure utilization metrics, and workload behavior. Instead of waiting until the monthly invoice arrives, organizations receive accurate forecasts that allow proactive financial planning and resource optimization.

The forecasting engine analyzes trends, seasonal variations, workload spikes, and historical spending patterns using advanced time-series machine learning models. This enables engineering and FinOps teams to anticipate budget overruns before they occur and take corrective actions early.

Features

- Daily cloud cost forecasting
- Weekly cloud cost forecasting
- Monthly cloud cost forecasting
- Quarterly expenditure prediction
- Annual cloud budget estimation
- Budget overrun prediction
- Seasonal workload forecasting

- Trend analysis and visualization
- Future expenditure estimation
- Cloud cost comparison reports

Benefits

- Prevent unexpected cloud bills
- Improve financial planning
- Optimize infrastructure proactively
- Enable data-driven budgeting
- Reduce unnecessary operational expenses

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2. ML-Based Unit Economics Forecasting

CloudSense AI extends beyond infrastructure monitoring by connecting cloud expenditure with business performance. Rather than simply reporting infrastructure costs, the platform evaluates how efficiently cloud resources contribute to business growth.

The system correlates cloud spending with organizational KPIs such as active users, API requests, customer growth, business transactions, and application sessions. This allows decision-makers to understand the true business value generated from cloud investments.

Unit Economics Examples

- Cost per Active User
- Cost per API Request
- Cost per Customer
- Cost per Transaction
- Cost per Application Session
- Cost per Tenant
- Cost per Microservice

Capabilities

- Business KPI correlation
- Cost-per-KPI forecasting
- Infrastructure scaling analysis

- Growth versus expenditure comparison
- Unit Economics visualization
- Business-aware cloud intelligence
- Financial efficiency measurement

Benefits

- Measure infrastructure efficiency
- Improve cloud ROI
- Support strategic business decisions
- Align FinOps with business objectives
- Forecast long-term operational costs

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3. Intelligent Resource Optimization

CloudSense AI continuously monitors cloud resources to identify inefficient utilization and optimization opportunities. Instead of relying on manual inspections, the AI engine automatically detects idle or oversized resources that increase cloud expenditure without delivering proportional business value.

The recommendation engine evaluates CPU utilization, memory consumption, storage utilization, network activity, workload patterns, and infrastructure dependencies before suggesting optimization strategies.

Detectable Resources

- Idle EC2 Instances
- Underutilized Compute Instances
- Over-Provisioned Virtual Machines
- Unattached EBS Volumes
- Obsolete Snapshots
- Idle RDS Databases
- Low-utilization Kubernetes Nodes
- Unused Elastic IP Addresses
- Inactive Load Balancers
- Underutilized Storage Buckets

Optimization Recommendations

- Instance Rightsizing
- Storage Optimization
- Resource Scheduling
- Automatic Shutdown Windows
- Reserved Instance Recommendations
- Savings Plan Recommendations
- Resource Consolidation
- Storage Tier Optimization
- Auto Scaling Improvements
- Lifecycle Policy Recommendations

Benefits

- Reduce cloud expenditure
- Increase resource utilization
- Improve infrastructure efficiency
- Simplify cloud management
- Reduce operational waste

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4. Denial-of-Wallet (DoW) Threat Detection

CloudSense AI integrates cybersecurity with FinOps by identifying Denial-of-Wallet attacks that intentionally increase cloud expenditure. Unlike traditional denial-of-service attacks that target service availability, DoW attacks silently consume cloud resources to create excessive operational costs.

The AI continuously analyzes cloud activity and compares current behavior with historical workload patterns to distinguish legitimate business growth from malicious cloud consumption.

Detection Strategy

The system analyzes:

- API request frequency
- Resource provisioning spikes
- Compute utilization
- Storage consumption
- Cost accumulation rate
- Network traffic behavior
- Infrastructure scaling patterns
- User access patterns
- Geographic request distribution
- Service usage anomalies

AI Classification

The system distinguishes between:

- Normal workload growth
- Seasonal traffic increases
- Marketing campaign traffic
- Business expansion
- Infrastructure failures
- Suspicious cloud activity
- Denial-of-Wallet attacks

Benefits

- Detect financial cyberattacks
- Prevent unnecessary cloud spending
- Protect cloud budgets
- Improve cloud security
- Strengthen FinOps and SecOps integration

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5. Explainable AI Recommendation Engine

One of the major challenges in AI-driven cloud optimization is building trust in automated recommendations. CloudSense AI addresses this challenge by integrating Explainable Artificial Intelligence (XAI), ensuring that every optimization decision is transparent, interpretable, and supported by meaningful evidence.

Instead of presenting only predictions, the platform explains why a recommendation was generated, what factors influenced the prediction, how much confidence the AI has in its decision, and the expected business impact after implementation. This enables cloud engineers, DevOps teams, and FinOps managers to confidently approve optimization actions.

The Explainable AI engine evaluates multiple cloud metrics such as CPU utilization, memory usage, storage consumption, network traffic, historical billing trends, workload behavior, business KPIs, and anomaly detection scores before generating optimization recommendations.

Each recommendation is accompanied by a detailed explanation that improves transparency and simplifies engineering decision-making.

Each Recommendation Includes

- Root Cause Analysis
- AI Confidence Score
- Prediction Explanation
- Feature Importance
- Estimated Monthly Savings
- Estimated Annual Savings
- Risk Level
- Business Impact
- Recommended Optimization Action
- Expected Performance Improvement
- Estimated Return on Investment (ROI)

Benefits

- Builds trust in AI recommendations
- Improves decision-making
- Reduces manual investigation
- Enables transparent cloud governance

- Simplifies FinOps operations
- Supports enterprise compliance

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6. Intelligent Cloud Remediation using MCP and AWS SDK (Boto3)

CloudSense AI transforms optimization recommendations into intelligent cloud operations through the Model Context Protocol (MCP). Instead of generating Infrastructure-as-Code templates or AWS CLI commands, the platform securely communicates with cloud environments using MCP while AWS SDK (Boto3) performs the approved remediation actions.

Every optimization recommendation follows a controlled workflow to ensure operational safety. After the AI identifies an optimization opportunity, the recommendation is reviewed by the engineering team. Once approved, MCP establishes communication with AWS services and Boto3 executes the requested cloud operation.

This architecture eliminates the complexity of Infrastructure-as-Code generation while providing secure, flexible, and enterprise-ready automation.

Supported Remediation Actions

Compute Optimization

- Start EC2 Instances
- Stop Idle EC2 Instances
- Restart EC2 Instances
- Resize EC2 Instances
- Change Instance Types
- Enable Auto Scaling
- Modify Auto Scaling Groups

Storage Optimization

- Delete Unattached EBS Volumes
- Remove Old Snapshots
- Optimize Storage Classes
- Archive Inactive Storage

- Configure Lifecycle Policies

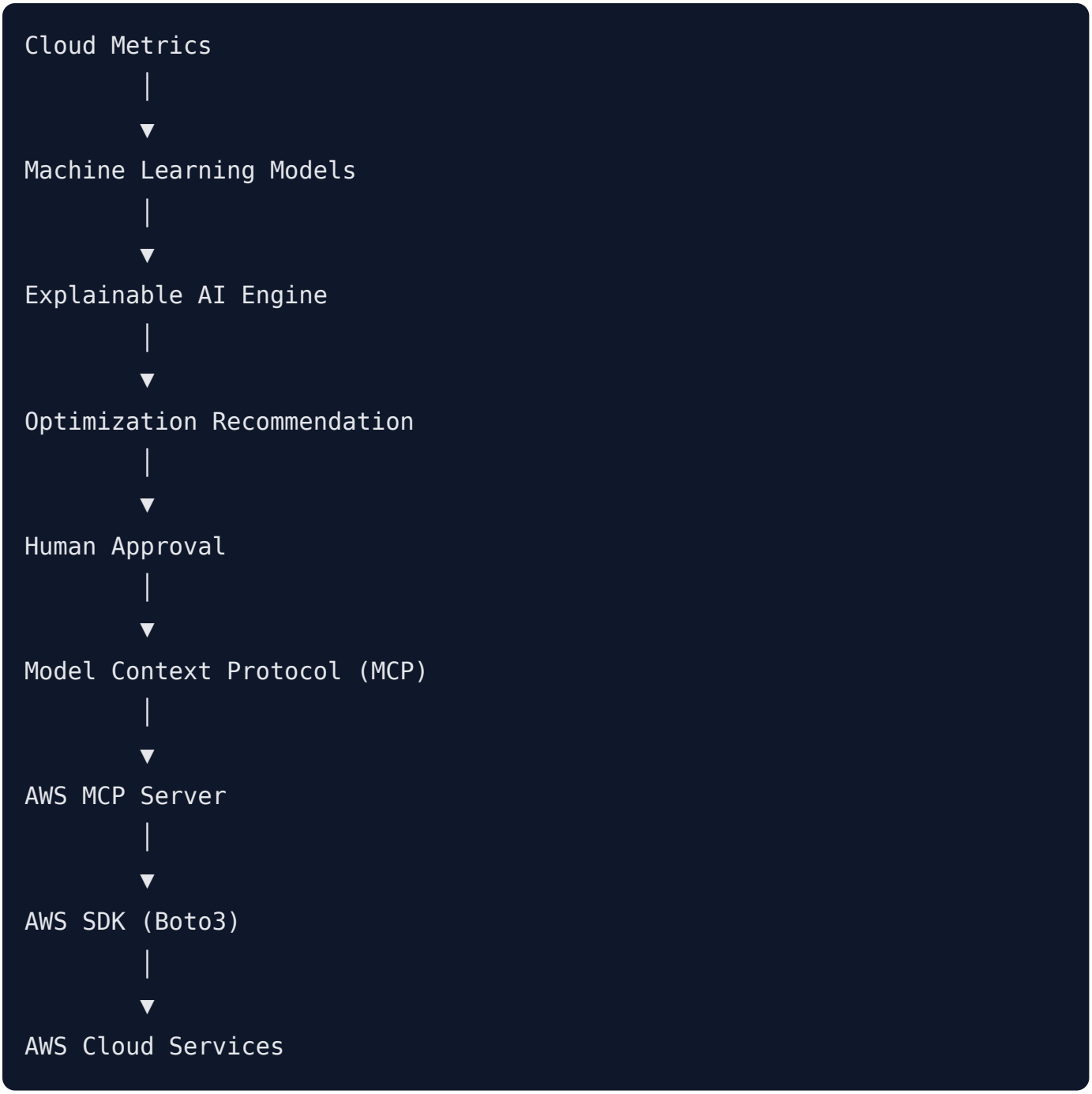
Database Optimization

- Start RDS Instances
- Stop RDS Instances
- Resize Database Instances
- Optimize Storage Allocation

Monitoring & Automation

- Configure CloudWatch Alarms
- Invoke AWS Lambda Functions
- Trigger Automated Workflows
- Generate Optimization Reports

Intelligent Workflow



Benefits

- Eliminates manual remediation
- Maintains human approval
- Secure enterprise automation
- Easy cloud integration
- Faster optimization
- Reduced operational effort
- Improved cloud governance

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7. Real-Time Intelligent Alerting

CloudSense AI continuously monitors cloud infrastructure and immediately informs engineering teams whenever abnormal cloud activity or optimization opportunities are detected.

Instead of waiting for engineers to manually review dashboards, the platform delivers actionable notifications directly to collaboration platforms used by development and operations teams.

Each alert includes sufficient context for engineers to quickly understand the issue and make informed decisions without manually investigating cloud metrics.

Supported Integrations

- Slack
- Microsoft Teams
- Discord
- Email
- Custom Webhooks
- Incident Management Platforms

Alert Information

Every notification includes:

- Cloud Cost Anomaly
- Root Cause
- AI Confidence Score
- Estimated Savings
- Risk Level
- Resource Details
- Recommended Action
- Priority Level
- Business Impact

Benefits

- Faster incident response

- Improved collaboration
- Reduced monitoring effort
- Immediate optimization opportunities
- Better operational visibility

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8. Interactive What-If Simulator

Before implementing cloud optimization recommendations, organizations often need to understand how proposed changes will affect infrastructure performance, operational costs, and business objectives.

CloudSense AI provides an Interactive What-If Simulator that allows engineers to evaluate optimization strategies in a safe virtual environment before applying changes to production systems.

The simulator estimates financial savings, predicts infrastructure utilization, evaluates workload performance, and compares multiple optimization scenarios using AI-based forecasting models.

Simulation Capabilities

- Current vs Optimized Cost Comparison
- Monthly Savings Estimation
- Annual Savings Estimation
- Resource Utilization Prediction
- Budget Impact Analysis
- ROI Calculation
- Performance Impact Prediction
- Capacity Planning
- Infrastructure Scaling Analysis
- Cost-Benefit Comparison

Benefits

- No production risk
- Better planning
- Improved financial forecasting

- Confident decision-making
- Higher optimization success rate

Technology Stack

CloudSense AI is built using a modern, scalable, and enterprise-ready technology stack that combines Artificial Intelligence, Machine Learning, Cloud Computing, Explainable AI, and modern cloud automation technologies. The selected technologies ensure high performance, scalability, maintainability, and seamless integration with AWS cloud services.

Backend

The backend is responsible for data collection, machine learning inference, API development, cloud orchestration, and communication with AWS services.

Technologies

- Python
- FastAPI
- Pydantic
- SQLAlchemy
- REST APIs

Responsibilities

- Cloud metric collection
- Cost analysis
- Machine learning inference
- Recommendation generation
- MCP orchestration
- Boto3 cloud execution
- Notification management

Frontend

The frontend provides an intuitive dashboard for engineers, FinOps analysts, and cloud administrators to visualize cloud spending, optimization opportunities, and AI recommendations.

Technologies

- React.js
- Tailwind CSS
- TypeScript
- Chart.js / Plotly

Features

- Cloud cost dashboard
- Forecast visualization
- Unit Economics dashboard
- Resource optimization reports
- Recommendation approval interface
- Alert management
- What-If Simulator

Database

The database securely stores cloud metrics, billing information, historical predictions, AI recommendations, user approvals, and optimization history.

Technology

- PostgreSQL

Stored Data

- Billing history
- Cloud metrics
- AI predictions
- Resource inventory
- Optimization logs
- User approvals

- Notification history

Cloud Integration & Automation

Instead of Infrastructure-as-Code, CloudSense AI uses the **Model Context Protocol (MCP)** together with **AWS SDK (Boto3)** for secure and controlled cloud remediation.

Technologies

- Docker
- Model Context Protocol (MCP)
- AWS SDK (Boto3)

Responsibilities

- Secure AWS communication
- Cloud remediation
- Resource management
- Workflow automation
- Notification routing

Machine Learning

The AI engine uses multiple specialized models to solve different cloud optimization tasks.

Libraries

- XGBoost
- Prophet
- Isolation Forest
- Scikit-learn
- Pandas
- NumPy

Visualization & Monitoring

CloudSense AI provides interactive dashboards that simplify cloud monitoring and decision-making.

Tools

- Plotly
- Grafana

Notifications

Engineering teams receive real-time notifications through multiple collaboration platforms.

Supported Platforms

- Slack
- Microsoft Teams
- Discord
- Email
- Custom Webhooks

Machine Learning Models

CloudSense AI combines multiple machine learning models, with each model solving a specific optimization problem.

Model	Purpose
Prophet	Cloud cost forecasting
XGBoost	Unit Economics prediction
Isolation Forest	Cloud anomaly and Denial-of-Wallet detection
Scikit-learn	Data preprocessing, feature engineering, model evaluation

Prophet

Used for forecasting future cloud expenditure based on historical billing trends, seasonal workload patterns, and long-term infrastructure growth.

XGBoost

Predicts business-aware Unit Economics and estimates cloud spending relative to organizational KPIs such as users, API requests, and transactions.

Isolation Forest

Detects abnormal cloud behavior, unusual spending spikes, idle resources, and Denial-of-Wallet attacks.

Scikit-learn

Supports preprocessing, feature engineering, model evaluation, and pipeline integration.

Explainable AI Components

CloudSense AI incorporates Explainable Artificial Intelligence (XAI) to ensure every recommendation is transparent and understandable.

Every AI recommendation includes:

- Prediction Explanation
- Root Cause Analysis
- Confidence Score
- Feature Importance
- Estimated Monthly Savings
- Estimated Annual Savings
- Risk Level
- Business Justification
- Recommended Action
- Expected Business Impact

Benefits

- Transparent AI
- Better engineering decisions
- Increased operational trust
- Easier compliance

- Human-centered optimization

Unique Contributions

CloudSense AI introduces several innovations that distinguish it from traditional cloud monitoring platforms.

Intelligent FinOps

Predicts future cloud expenditure instead of simply reporting historical costs.

Explainable AI

Every recommendation is supported with detailed reasoning and confidence scores.

FinOps + SecOps Integration

Combines financial optimization with cloud security to detect Denial-of-Wallet attacks.

Business-Aware Cloud Intelligence

Correlates cloud spending with organizational KPIs to produce meaningful Unit Economics.

MCP-Based Cloud Orchestration

Uses Model Context Protocol to securely communicate with enterprise cloud services.

Intelligent Cloud Remediation

Uses AWS SDK (Boto3) to execute approved optimization actions instead of Infrastructure-as-Code templates.

Evaluation Metrics

The performance of CloudSense AI is evaluated using multiple business, AI, and operational metrics.

Category	Evaluation Metric	Target

Cost Forecasting	Mean Absolute Percentage Error (MAPE)	< 5%
Unit Economics	Mean Absolute Error (MAE)	Minimize
DoW Detection	Precision	High
DoW Detection	Recall	High
DoW Detection	F1-Score	High
Optimization	Estimated Monthly Savings	Maximum
Recommendation Quality	AI Confidence Score	High
API Performance	Response Time	Low Latency
Cloud Automation	Remediation Success Rate	High Reliability

Expected Outcomes

After deployment, CloudSense AI is expected to deliver measurable improvements across cloud operations, security, and financial management.

Expected outcomes include:

- Significant reduction in cloud expenditure
- Higher infrastructure utilization
- Accurate cloud cost forecasting
- Early budget overrun detection
- Real-time Denial-of-Wallet detection
- Intelligent cloud remediation
- Reduced manual operational effort
- Improved FinOps decision-making
- Business-aware Unit Economics
- Explainable AI-driven optimization
- Enterprise cloud governance
- Faster engineering response
- Improved cloud sustainability

Why This Project Stands Out

CloudSense AI is a production-ready enterprise platform that combines Artificial Intelligence, Machine Learning, Cloud Computing, FinOps, SecOps, Explainable AI, Model Context Protocol (MCP), and AWS SDK (Boto3) into a single intelligent cloud optimization solution.

Unlike conventional cloud monitoring tools that only visualize historical spending, CloudSense AI predicts future infrastructure costs, explains AI decisions, correlates cloud expenditure with business performance, detects financial cyber threats such as Denial-of-Wallet attacks, and securely executes approved cloud remediation through MCP and Boto3.

The project demonstrates expertise across multiple high-demand domains including Artificial Intelligence, Machine Learning, Predictive Analytics, Cloud Engineering, FinOps, Cloud Security, Explainable AI, Backend Development, Full-Stack Development, Enterprise Automation, and Intelligent Decision Support Systems.

Its practical business value, enterprise architecture, AI-driven decision-making, and modern cloud automation approach make it highly suitable for IEEE research publications, capstone projects, hackathons, internships, and campus placements.

Conclusion

CloudSense AI represents a next-generation intelligent cloud optimization platform that combines predictive analytics, explainable machine learning, financial intelligence, cloud security, and enterprise automation into one integrated solution.

By forecasting cloud expenditure, optimizing infrastructure, detecting Denial-of-Wallet attacks, and securely executing approved remediation through Model Context Protocol (MCP) and AWS SDK (Boto3), the platform enables organizations to reduce costs, improve resource utilization, strengthen cloud governance, and make data-driven operational decisions with confidence.

CloudSense AI bridges the gap between FinOps, DevOps, and SecOps, delivering an intelligent, explainable, and scalable cloud management framework suitable for modern enterprise environments.